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TinkerWithMe

Self-Driving Car · Premium Course

3 live sessions · 2 hrs each · 11–15 yrs · from KES 9,000

Premium course outline **PREMIUM**

Overview

Build a chassis-based car that navigates on its own, finishing with a race day.

DETAIL	
Track	Arduino Robotics & Electronics
Format	3 live sessions · 2 hrs each
Delivery	Live online (1:1 or small group) or in person, you choose when booking
Ages	11–15 yrs
Level	Intermediate → Advanced
Price	from KES 9,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16×2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Needed at home for live online sessions; for in-person sessions we provide kits, so buy one only to keep building at home.
Parts pack	Optional, KES 2,500: 2WD car chassis with 2 DC motors and wheels, L298N motor driver, Second IR sensor, AA battery pack
Standalone	No
Prerequisites	Servo motor control (free)

Programme

1. **Motorised Chassis.** Assemble the chassis and wire DC motors to an L298N motor driver, then program forward, reverse and turn controls and drive a test course.

2. **Line Follower.** Calibrate IR sensors on black vs. white, then program the car to follow a track on its own and tune its speed on the curves.
3. **Obstacle Avoidance.** Mount an HC-SR04 ultrasonic sensor so the car detects and steers around objects, then race the finished cars on race day!

What they take home

🎁 Working self-driving car kit

Why parents love it

“The technology behind Waymo and Tesla — built from scratch by your 12-year-old.”

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**Self-Driving Car**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.